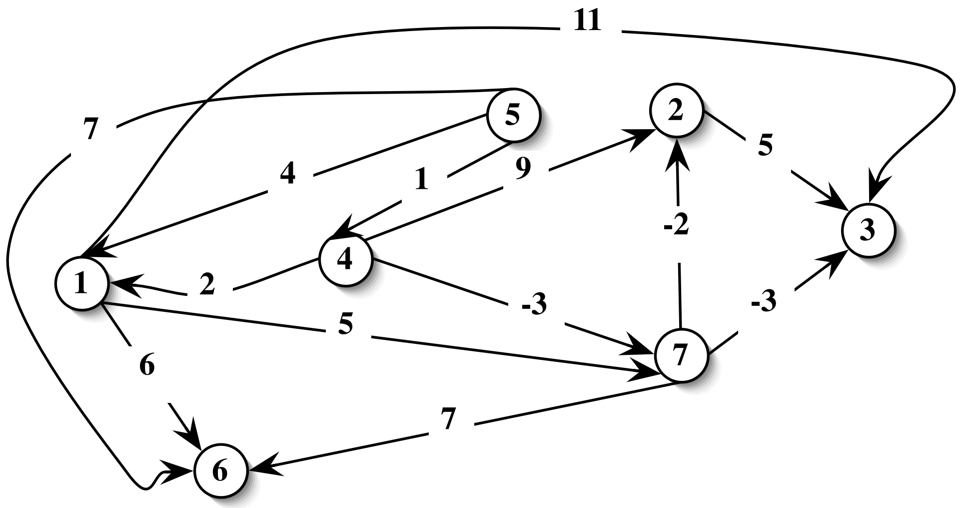
First part : Closed book

Total score: 17 pts. Minimum score to pass: 12 pts.

1. Consider the following directed graph. Find the shortest path with root in node 5. Apply the most efficient algorithm, after a possible preprocessing phase. Justify your choice and illustrate the intermediate steps. What is the complexity of the algorithm?



2. Consider the following LP problem:

$$\begin{aligned} \min \quad & 5x_1 + 2x_2 + 7x_3 + 3x_4 \\ & -2x_1 + 4x_2 \quad - x_4 = 3 \\ & 2x_1 + 2x_2 - 2x_3 - 2x_4 = -1 \\ & x_i \geq 0 \quad i = 1, \ldots, 4 \end{aligned}$$

Write the dual and complementary slackness equations. Check the optimality for the following solutions

$$x' = [0, 7/6, 0, 10/6] \qquad x'' = [0, 3/4, 5/4, 0]$$

3. Consider the following ILP problem:

$$\begin{aligned} \min \quad & -x_1 + 4x_2 \\ & 4x_1 + 6x_2 \leq 27 \\ & 2x_1 - 6x_2 \leq 9 \\ & x_1, x_2 \geq 0 \end{aligned}$$

After adding suitable slack variables, consider the optimal base $B = \{1, 3\}$ of the LP relaxation. Compute the Gomory cut(s) with respect to the corresponding solution. Give the geometrical representation of the found cut(s).

4. Consider the minimum spanning tree problem. Write the decision version of the problem.